

Rover CAN Message Format

Jetson (onboard computer) <-> Teensy 4.1 (embedded controller) | CAN 2.0B, 11-bit IDs, 500 kbps, little-endian



CAN IDs are priorities: the lowest ID wins the bus

Arbitration is bitwise and dominant-low. The losing node backs off and retries; its frame is not corrupted.

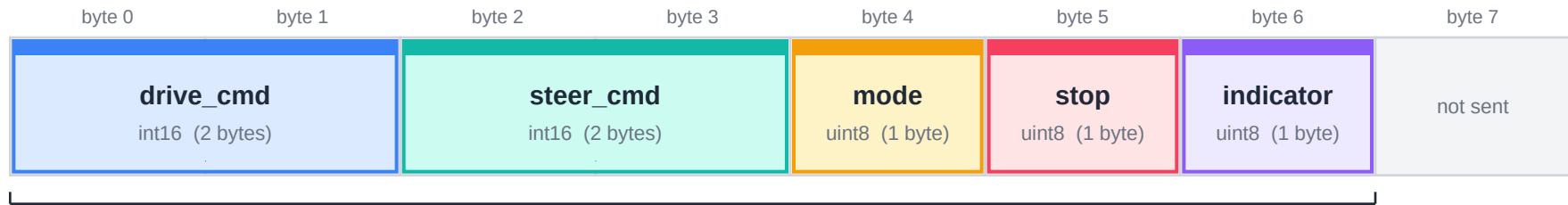
CAN ID	Message	Direction	DLC	Rate	Priority
0x000-0x0FF	reserved (future dedicated e-stop)	-	-	-	highest
0x100	CONTROL	Jetson -> Teensy	7	20 Hz	high
0x200	TELEM_MOTION	Teensy -> Jetson	8	20 Hz	low
0x201	TELEM_STATUS	Teensy -> Jetson	8	20 Hz	lowest

Link basics

- Every message fits in 8 bytes, so it runs on Classic CAN (CAN1/CAN2) or CAN FD (CAN3) unchanged.
- CAN hardware provides framing, a 15-bit CRC and automatic retransmission - there is no software CRC.
- The receiver checks the DLC (length). A wrong DLC is rejected and does NOT refresh the command watchdog.
- Multi-byte fields are little-endian: lowest byte first. Example: 300 = 0x012C is sent as 2C 01.
- 8-byte frame at 500 kbps = about 220 us on the wire, about 0.4% of the 50 ms period.
- All of these values live in firmware/src/rover_config.h and host/rover_config.py (kept in sync by a test).

CONTROL 0x100 DLC 7

Jetson -> Teensy, 20 Hz. Also acts as the heartbeat: no valid CONTROL for 300 ms = rover stops.



Example payload (hex, little-endian): drive=300, steer=-200, mode=MANUAL, stop=0, indicator=BLUE

2C	01	38	FF	01	00	01	--
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Field	Bytes	Type	Valid range	Meaning
drive_cmd	0-1	int16	-1000 .. 1000	Drive effort in tenths of a percent (1000 = 100% forward)
steer_cmd	2-3	int16	-1000 .. 1000	Steering in tenths of a percent of full range
mode	4	uint8	0, 1, 2	0 = DISABLED, 1 = MANUAL, 2 = AUTONOMOUS
stop	5	uint8	0, 1	1 = stop immediately, regardless of mode
indicator_request	6	uint8	0, 1, 2, 3	Status light the Jetson asks for: 0 OFF, 1 BLUE, 2 RED, 3 GREEN_FLASH

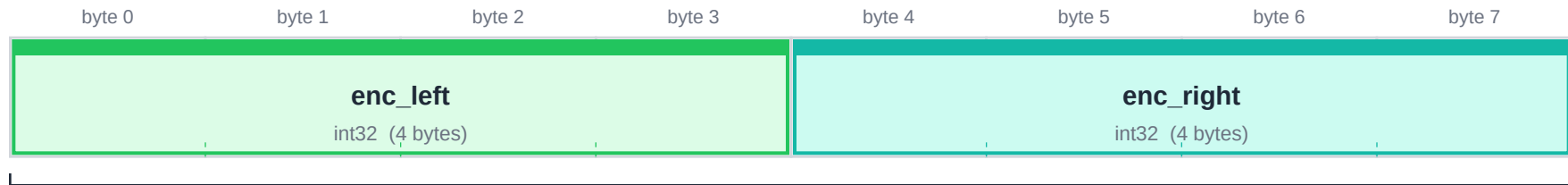
Validation: a frame is REJECTED (not clamped) if drive/steer is outside -1000..1000, or mode or indicator_request is undefined.

A rejected frame does not refresh the 300 ms watchdog, so the rover stops, and TELEM_STATUS reports PROTOCOL_ERROR (0x40).

Telemetry 0x200 + 0x201

Teensy -> Jetson, 20 Hz each. Split into two frames so each fits Classic CAN's 8-byte limit.

TELEM_MOTION 0x200 DLC 8



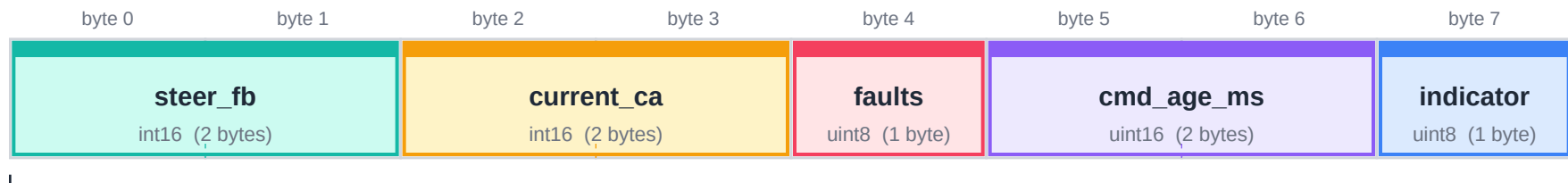
DLC = 8 bytes on the wire

Example payload (hex, little-endian): enc_left = enc_right = 36550

C6	8E	00	00	C6	8E	00	00
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Cumulative encoder ticks. They wrap around at the int32 limit, so only use differences between readings.

TELEM_STATUS 0x201 DLC 8



DLC = 8 bytes on the wire

Example payload (hex, little-endian): steer_fb=-150, current=4.50 A, no faults, cmd_age=42 ms, indicator=BLUE

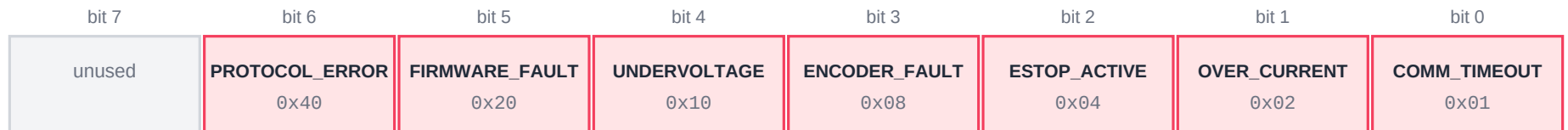
6A	FF	C2	01	00	2A	00	01
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TELEM_STATUS field details

Fault bitmask and the reserved command-age values

Field	Bytes	Type	Range	Meaning
steer_fb	0-1	int16	-1000 .. 1000	Measured steering, same scale as steer_cmd
current_ca	2-3	int16	±327.67 A	Signed current in centiamps (1 cA = 10 mA). Negative while braking.
fault_status	4	uint8	bitmask	Faults OR'ed together - see the bit layout below
cmd_age_ms	5-6	uint16	0 .. 65534	Milliseconds since the last valid CONTROL frame
indicator_state	7	uint8	0, 1, 2, 3	Status light the Teensy is ACTUALLY showing (may differ from the request)

fault_status bit layout (byte 4)



COMM_TIMEOUT - no valid CONTROL within 300 ms (rover is stopped)

OVER_CURRENT - current exceeded the threshold

ESTOP_ACTIVE - hardware or software e-stop engaged

ENCODER_FAULT - encoder reading invalid or stalled

UNDERVOLTAGE - supply voltage low (placeholder)

FIRMWARE_FAULT - this boot followed a hardware-watchdog reset

PROTOCOL_ERROR - last CONTROL was rejected: wrong DLC, undefined mode/indicator, or drive/steer out of range

Example: COMM_TIMEOUT + OVER_CURRENT = 0x01 | 0x02 = 0x03

cmd_age_ms reserved values

Value	Name	Meaning
0xFFFF	CMD_AGE_UNKNOWN	No valid CONTROL has EVER arrived - check wiring / bus config
0xFFFE	CMD_AGE_MAX	Real measurement saturated - Jetson stopped talking 65+ s ago